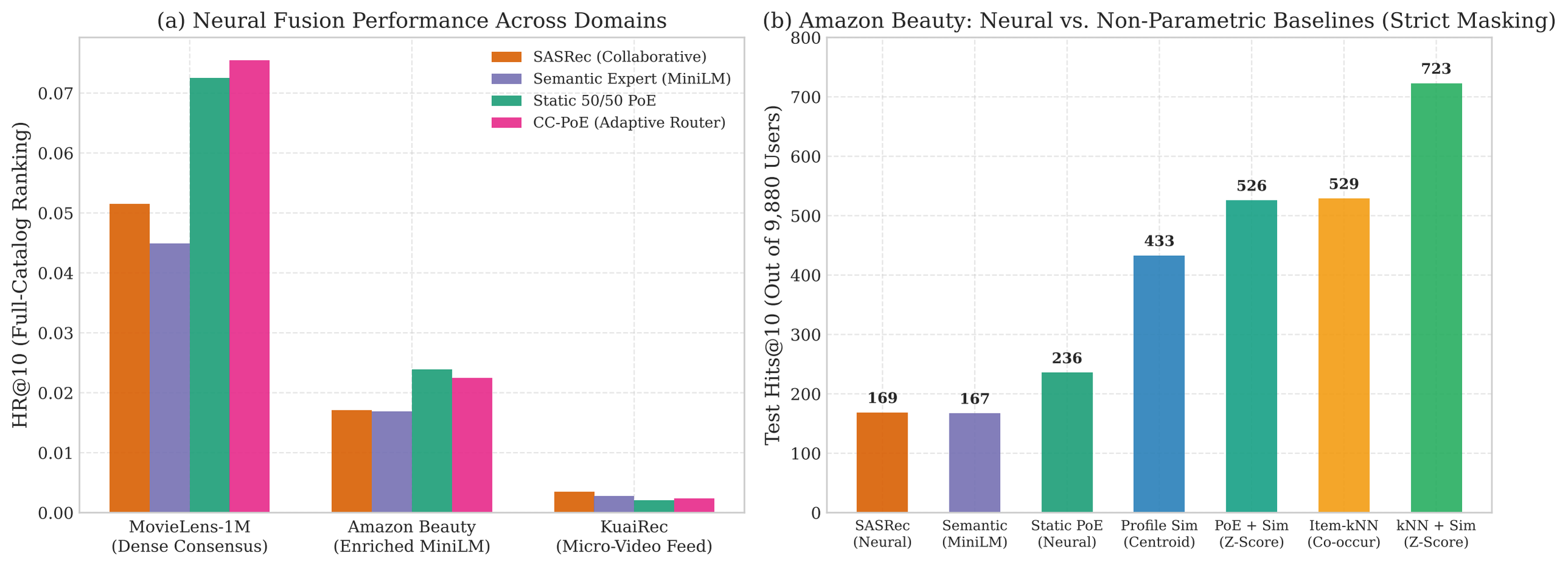
(a) Neural Fusion Performance Across Domains
HR@10 (Full-Catalog Ranking)
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
SASRec (Collaborative)
Semantic Expert (MiniLM)
Static 50/50 PoE
CC-PoE (Adaptive Router)
MovieLens-1M
(Dense Consensus)
Amazon Beauty
(Enriched MiniLM)
KuaiRec
(Micro-Video Feed)
(b) Amazon Beauty: Neural vs. Non-Parametric Baselines (Strict Masking)
Test Hits@10 (Out of 9,880 Users)
0
100
200
300
400
500
600
700
800
169
167
236
433
526
529
723
SASRec
(Neural)
Semantic
(MiniLM)
Static PoE
(Neural)
Profile Sim
(Centroid)
PoE + Sim
(Z-Score)
Item-kNN
(Co-occur)
kNN + Sim
(Z-Score)